AI-Assisted Debugging & Problem Solving

Alkem MedTech Learning Lab · A practical example from production engineering

My role Full Stack Engineer	AI tools Cursor Agent + ChatGPT	Focus Debugging, investigation & implementation

1. The Problem

The Alkem platform uses encryption middleware for sensitive data and uploaded media. During file uploads, the encryption flow converted the file data to Base64. Smaller files worked as expected, but larger files failed with a “Maximum call stack size exceeded” error.

2. How I Investigated It

I first reproduced and analysed the failure, then traced the upload and encryption flow to understand where the large payload was being processed. I used Cursor Agent alongside my own debugging to explore the relevant code, follow the data flow and reason about possible causes. The investigation pointed to the way the complete file was being handled during the Base64 conversion/encryption process.

3. Where AI Helped

• Codebase exploration: used Cursor Agent to navigate the existing upload and encryption flow and identify related files and functions.

• Problem analysis: used AI to discuss possible causes of the stack-size failure and compare implementation approaches.

• Implementation support: used Cursor Agent while iterating on the chunk-based processing change.

• Validation: reviewed the generated changes, tested the behaviour with larger files, and made the final engineering decisions myself.

4. The Solution

The original approach attempted to process the entire file at once. I changed the flow to process the file in smaller chunks, convert the chunks to Base64 as part of the encryption workflow, and continue processing without trying to handle the complete large payload in one operation.

Before	**After**
Large file → process entire payload → Base64/encryption flow → stack-size failure	Large file → process in chunks → Base64/encryption flow → continue processing

5. Result

The upload/encryption flow was changed to use chunk-based processing instead of attempting to process the entire large file at once, addressing the failure encountered with larger uploads.

6. What This Shows About My AI Workflow

I use AI to help me get through problems faster, especially when I need to understand an unfamiliar part of a codebase or look at a problem from a different angle. In this case, Cursor helped me trace the issue and explore possible fixes, but I still had to understand the problem, test the changes and decide what should actually go into the application. I see AI as a tool that helps me think and work faster, not something that replaces the engineering work.